Unique id: E0224033
Name: Swaminathan

Project Title: Proactive Water Quality Deterioration Prediction Using IoT Data and Machine Learning

Dataset Description:
This dataset contains 100,800 in-situ physicochemical water quality observations captured using an IoT-based multi-sensor telemetry system deployed across diverse natural and treated water sources in Kenya during February–March 2026. The measurements were recorded continuously at a 2-minute sampling interval across 10 distinct 14-day monitoring campaigns (10,080 readings per deployment). The system monitored four key physicochemical parameters: pH, Total Dissolved Solids (TDS in ppm), Turbidity (in NTU), and Temperature (in °C). The dataset covers four contrasting water source categories: treated municipal and commercial bottled water (Safe), surface water from rivers and lakes subject to silt and runoff (Unsafe), mineral-rich groundwater from boreholes (Unsafe), and coastal marine water from the Indian Ocean (Unsafe). Each record includes a binary potability label ('Safe' vs. 'Unsafe') established against World Health Organization (WHO) and Kenyan national drinking water quality standards.

Dataset Download Link: https://data.mendeley.com/datasets/575ctg3nnr/2
Repository: Mendeley Data (Elsevier) | DOI: 10.17632/575ctg3nnr.2 (Version 2, April 2026)

Data Dictionary:
Shape of the Data:
No of Rows: 100,800
No of Columns: 5 (4 features + 1 target label)
Total Storage Size: 2.83 MB

Features (Variable)	Type Variable	Description
pH	Continuous Numerical (float64)	Logarithmic measure of hydrogen-ion concentration indicating water acidity or alkalinity (0–14 scale). WHO potable drinking water guidelines recommend a target range of 6.5 to 8.5.
TDS_ppm	Continuous Numerical (float64)	Total Dissolved Solids measured in parts per million (ppm = mg/L), quantifying aggregate dissolved inorganic salts and minerals. The desirable drinking water threshold is below 300–500 ppm.

Turbidity_NTU	Continuous Numerical (float64)	Nephelometric Turbidity Units measuring light scattering caused by suspended sediments, silt, and colloidal matter. The standard potable threshold is below 5.0 NTU (ideally < 1.0 NTU).
Temperature_C	Continuous Numerical (float64)	Ambient water temperature in degrees Celsius (°C). Influences chemical dissolution equilibria, microbial reaction kinetics, and electrical conductivity calibration of sensors.
Label (Target)	Binary Categorical (object / string)	Ground-truth water potability classification ('Safe' vs. 'Unsafe'). 'Safe' represents treated municipal and bottled water (30.0%); 'Unsafe' represents untreated surface, borehole, and marine sources (70.0%).
pH_change / TDS_change	Continuous Numerical (float64, Engineered)	First-order rate of change of sensor values over the preceding observation window (e.g., TDS_t - TDS_t-k), capturing dynamic deterioration trends and sudden runoff spikes.
rolling_mean / rolling_std	Continuous Numerical (float64, Engineered)	Rolling window statistical features (computed over 5 to 10 observations = 10–20 min), quantifying short-term sensor volatility and progressive contamination momentum.
unsafe_in_next_20_minutes	Binary Categorical (int64, Engineered Target)	Proactive early-warning prediction target indicating whether water will deteriorate into an unsafe condition 20 minutes into the future (enabling automated preventative valve actuation).